Xiangbo Gao

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Research Intern: Autonomous Driving | Perception | Motion Planning

EDUCATION

BS, University of California, Irvine Computer Science, Mathematics (GPA-3.720) 2018.9 ~ 2023.3

Summer Session, University of California, Berkeley (GPA-3.566) 2019.6 ~ 2019.9

SKILLS & Research Interest

Programming Language: Python, C/C++ | Software: PyTorch, OpenMMlab, Carla Simulator, RLlib, TorchGeometry

Research Interests: Machine/Deep Learning, Computer Vision, Perception, Motion Prediction, and Planning.

PUBLICATIONS

1. **X Gao**[1], et al. “Scale-free and Task-agnostic Attack: Generating Photo-realistic Adversarial Patterns with Patch Quilting Generator” arXiv:2208.06222, 2022 (On Submission)
2. M Liu, X Li, **X Gao**[3], J Chen, L Shen, & H Wu "Sample hardness based gradient loss for long-tailed cervical cell detection", International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 109-119. Springer, Cham, 2022

RESEARCH EXPERIENCE & LEADERSHIP

Auto-generated Graphical Model in the Autonomous Driving System 2022.5 ~ Present

- Designed a multi-domain autonomous driving dataset for various driving scenarios using the CARLA simulator.
- Developed a probabilistic LSTM structure that encodes states as variational embeddings using Pytorch.
- Evaluating and comparing the model performance and cross-domain transferability.

Multi-modal 3D Object Detection in autonomous driving scenario 2022.6 ~ Present

- Conducted a literature review of existing general 3D object detection algorithms, including camera-based, point-based, voxel-based, and multi-modal algorithms.
- Reproduce and improve existing multimodal 3D object detection algorithms using OpenMMlab.

Goal-conditional Reinforcement Learning 2022.2 ~ 2022.6

- Proposed a Spring Loss that uses the idea of contrastive learning to align the embeddings in the linear distance. Visualized the learned embedding using PCA, which shows better alignment with real-world trajectories.

Adversarial Attack with Semantic Pattern 2021.4 ~ 2022.10

- Proposed a novel Patch Quilting Generative Adversarial Network (PQ-GAN) that generates scale-free adversarial perturbation and deliveries state-of-the-art attack strength and robustness

Long-tailed Cervical Cell Detection 2021.5 ~ 2022.1

- Conducted experiments and contributed to 20% of paper writing for a proposed Grad-Libra Loss that uses gradients to dynamically calibrate sample hardness and rebalance gradients.
- Experiments show that our loss function achieved 7.8% higher mAP than CE loss on the long-tailed TCT WSI dataset.

ZerO Waste Anteaters 2020.9 ~ 2022.4

- Led a team of 8 members to train and deploy lightweight waste image classification and object detection models.
- Designed a small Image Recognition tutorial with side projects to help interested students get started in this field.

INTERNS & EMPLOYMENTS

Tandll Investment Management Limited, Shenzhen, China 2020.6 ~ 2020.8

Position: Full-stack Software developer | Main contribution: Management website currently in used

Calit 2, University of California, Irvine 2019.2 ~ 2019.7

Position: Software developer | Main contribution: VR teaching tools

COMPETITIONS

Netease Hackathon Competition 2020.6

Outstanding Award: A website for real-time traffic statistical monitoring using React, GraphQL, Rest, and MongoDB

UCI Machine Learning Hackathon, University of California, Irvine, CA 2020.4

(1st place) Geometric Pose Affordance: predicting 3D pose from 2D data and light tracing information

Google Hash Code 2020 Algorithms Competition 2020.2

2nd place / 13 at UCI